

Denoising diffusion probabilistic model-enabled data augmentation method for intelligent machine fault diagnosis

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ABSTRACT

Bearing fault is one of the main causes of rotating machinery failure. However, collecting sufficient failure data proves challenging in real-world industrial environments due to their complexity. Owing to this constraint, the majority of current methods fail to accurately identify fault types with limited data, thereby impeding timely maintenance efforts. To address this issue, we propose a bearing fault diagnosis method utilizing diffusion model data enhancement in this study. Following data collection, we employ the continuous wavelet transform to convert various one-dimensional vibration data into two-dimensional time series graphs. Subsequently, these feature graphs are partitioned into sets of fault samples. Then, the diffusion model is employed to augment the training samples. These augmented samples are then fed into the convolutional neural network for fault diagnosis, and their diagnostic accuracy is compared with that of the original dataset. The comparative analysis demonstrates that the data augmentation technique, founded on the diffusion model, enhances fault diagnosis accuracy within a small sample training dataset. Ultimately, the efficacy of the proposed method is validated utilizing the Paderborn bearing dataset and juxtaposed against alternative data augmentation techniques. The findings indicate that the diffusion model-based data augmentation method outperforms other techniques in terms of both training accuracy and stability, particularly in scenarios with small sample.

1. Introduction

Rotary machinery has found extensive application in modern industrial production (Tang et al., 2024; Li et al., 2021; Zhou et al., 2019). Bearings, as essential components in rotating machinery, hold a pivotal position in the manufacturing process. Considering the challenging working environment and intricate structure, the primary components of the bearing—including the inner ring, outer ring, and rolling body—are susceptible to damage. As bearing failures can result in both economic losses and potential casualties, prognostics and health management (PHM) especially fault diagnosis is of great significance for rotating machinery in order to ensure reliability and safety (Wang et al., 2024; Kim et al., 2024; Shi et al., 2024; Yang et al., 2024a). In real-world engineering scenarios, fault diagnosis of bearings faces complex challenges, including intricate system structures, varying operating conditions, data noise interference, fault evolution, limited data availability, nonlinear relationships, and high costs and safety risks. These factors make accurate fault diagnosis difficult but essential for maintaining

system reliability and performance. Various signal processing methods have been effectively utilized in analyzing fault signals derived from vibration data (Li et al., 2015; Ragab et al., 2016; Shen et al., 2014), with exploration extending to other signals like acoustic emission. These methods are commonly categorized into three domains (Chen et al., 2015): time domain, frequency domain, and time-frequency domain (Wang et al., 2017; Wu et al., 2017). Traditional approaches typically necessitate substantial prior expertise in fault diagnosis and signal processing, rendering them less suitable for industrial contexts. Consequently, intelligent fault diagnosis methods, which yield reliable results without extensive professional knowledge prerequisites, are gaining traction. Extensive data-driven research has leveraged machine learning and statistical inference techniques, including artificial neural networks (ANN) (Bin et al., 2012; Tian, 2012), support vector machines (SVM) (Zhang et al., 2015), random forest (RF) (Yang et al., 2008), fuzzy reasoning, and other refined algorithms (Wang et al., 2017, 2019; Aydin et al., 2015; Zhang et al., 2013).

Among various bearing fault detection methods, intelligent fault

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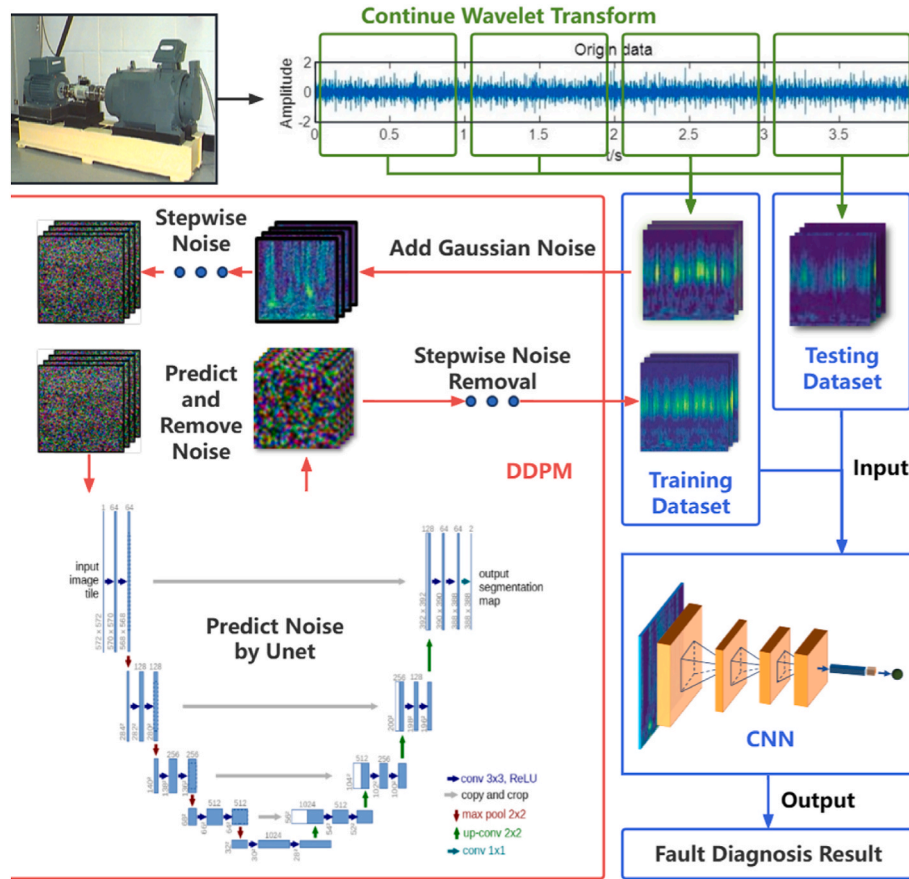


Fig. 1. The fundamental process diagram of data augmentation for DDPM-based bearing fault diagnosis.

diagnosis technology stands out as the most widely employed (Li et al., 2024a). With the swift advancement of artificial intelligence, PHM leveraging deep learning has emerged as a prominent research focus (Wang et al., 2024; Kim et al., 2024; Shi et al., 2024; Yang et al., 2024a; Song et al., 2024; Zhang and Hu, 2019; Yang et al., 2024b; Zhao and Chen, 2024; Qiang et al., 2024; Balshaw et al., 2024; Li et al., 2024b). Additionally, multi-modal data fusion and adaptive online monitoring technologies have improved both the comprehensiveness and real-time capabilities of fault detection (Ma et al., 2018; Peng et al., 2024; Zhang et al., 2024). Generative Adversarial Networks (GANs) are being used for data augmentation (Li et al., 2022), while reinforcement learning optimizes fault detection strategies (Fan et al., 2022; Li et al., 2023a). Feature selection and dimensionality reduction techniques have also increased model efficiency and interpretability (Zheng et al., 2024; Natsumeda and Yairi, 2024). These developments reflect ongoing innovation and progress in the field of machine fault diagnosis.

One-dimensional time series data in fault diagnosis often face challenges such as limited frequency domain information, difficulty extracting hidden features, high noise sensitivity, and poor interpretability. The Wavelet Transform (WT) offers a broader selection of basis functions to precisely match specific fault symptoms, facilitating efficient fault feature extraction. Converting vibration data into two-dimensional images enables the use of convolutional neural networks (CNNs), which are highly effective at detecting spatial patterns and features in images. This method can reveal complex patterns in the vibration data that may be challenging to identify in raw time-series format. Additionally, the two-dimensional image facilitates easier visualization and interpretation of data patterns, anomalies, and trends, highlighting crucial characteristics for accurate fault diagnosis. Representing vibration data as images allows for the application of advanced image processing and machine learning techniques. Utilizing pre-

trained image models and sophisticated algorithms tailored for image data can further enhance fault detection and diagnosis. Converting vibration data into two-dimensional images using Continuous Wavelet Transform (CWT) addresses these limitations by capturing time-frequency features, enabling multi-scale analysis, visually representing fault characteristics, and effectively separating noise. This approach enhances feature representation, improving the performance and reliability of fault diagnosis models. Additionally, the original generation model struggles to generate high-quality one-dimensional time series data. With the benefits of multi-resolution analysis, the practicality of wavelet transform in diagnosing faults in rotating machinery has been convincingly demonstrated (Chen et al., 2016).

In the presence of intricate noise within the sample and negligible distinctions among diverse fault characteristics, the transformation of the sample into two-dimensional imagery via the CWT approach may still preclude the explicit delineation of fault attributes. Consequently, this can result in the oversight of subtle disparities between normal and fault data, thereby manifesting as an issue of diminished sensitivity in detection. Nevertheless, deep learning techniques have addressed these challenges. Deep learning methods employ intricate multi-layered neural network architectures that facilitate the efficient handling and processing of substantial datasets, enabling the acquisition of sophisticated image features. They have been demonstrated to exhibit superior performance in terms of efficiency and precision when dealing with extensive volumes of data. Nonetheless, the requirement for copious amounts of data to facilitate training poses considerable difficulties for deep learning in practical, real-world scenarios. In actual industrial production, rolling bearing faults are frequently sporadic and rare. This leads to an imbalance in the types of available bearing faults for collection, and the sample data amount is consequently limited. The imbalanced distribution of data poses a challenge for traditional

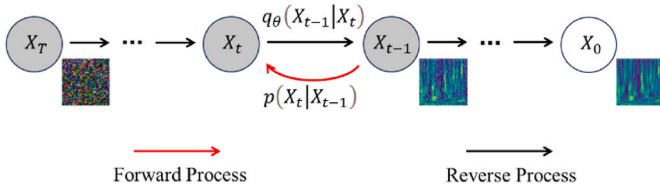


Fig. 2. DDPM processes.

methods (Li et al., 2023b), often leading to issues like overfitting, especially in the presence of a small sample fault database. This can adversely impact the effectiveness of the diagnosis process. In the case of issues like sample imbalance, traditional solutions have limitations in optimizing diagnostic performance (Lu et al., 2022). In recent years, there has been increasing research on the use of generative adversarial networks for data enhancement and fault diagnosis of small samples. Wang et al. proposed adaptive variational autoencoding generative adversarial networks (AVAEGAN) for data enhancement and application to fault diagnosis (Wang et al., 2023). Chen et al. proposed an innovative data augmentation approach for addressing the small-sample imbalanced problem: the balanced conditioning convolutional neural networks (CNN) Transformer architecture based generative adversarial network (BCTGAN) (Chen et al., 2024). Jian et al. proposed a novel open-set domain generalization (OSDG) learning framework that integrates a data generation module (DGM) and a feature learning module (FLM) (Jian et al., 2024). Li et al. proposed a novel fault diagnosis method using variational autoencoder generation adversarial network and improved convolutional neural network, named VGAIC-FDM (Li et al., 2024c). Zhang et al. presented a prior knowledge-augmented self-supervised feature learning framework for few-shot fault diagnosis (Zhang et al., 2022). Shen et al. presented a stable synthetic minority oversampling technique (SSMOTE) to generate the artificial fault data, which are then mixed with the raw fault data to achieve data augmentation (Shen et al., 2023). Jonathan Ho et al. (2020) introduced denoising diffusion probabilistic models (DDPM), a technique commonly employed in the domains of machine learning and image processing. This method is utilized for generating new samples or images. DDPM is grounded in a concept known as the “diffusion process,” encompassing two phases of noise-adding and denoising. In scenarios with limited bearing fault samples, denoising diffusion probabilistic models can generate high-quality and diverse fault data through its step-by-step generation and optimization process, robust denoising capability, and training stability. This is essential for expanding the training dataset in fault diagnosis tasks and plays a significant role in improving the accuracy and reliability of the diagnostic models. Further details on this process are elaborated in the subsequent sections.

The convolutional network leverages its feature extraction function to derive more precise features from matrix or spectral images of bearing fault data (Shao et al., 2021). While DDPM has achieved maturity in the field of image processing (Croitoru et al., 2023; Kazerouni et al., 2023), It aids in enhancing the quality of images by optimizing color, contrast, and resolution, thereby restoring their original characteristics (Guan et al., 2024; Lu et al., 2023). Additionally, it excels in eliminating unwanted occlusions or interference, thus refining the overall visual clarity (Zhang et al., 2023; Perera et al., 2023; Tao, 2024). Furthermore, the diffusion model can generate simulated images or data under specific conditions (He et al., 2024). In the realm of medical imaging, particularly in tasks like biomedical image segmentation, the diffusion model is also gradually applied (Liu et al., 2024). Multispectral images with high spatial resolution can be recovered by using diffusion model (Meng et al., 2023). The diffusion model can be used to generate images for data enhancement (Chen et al., 2023). There is a scarcity of research on the generation of two-dimensional images of fault data and the expansion of the sample database. In this paper, we delve into sample augmentation for mechanical vibration data. In the case of sample-based

augmentation, we assume that the training and testing input samples are derived from images obtained through the continuous wavelet transform of the original dataset. The diffusion model is then employed to augment the training set directly. The fundamental process diagram is illustrated in Fig. 1.

The remainder of this article initiates with an explanation of the proposed approach in the “Proposed Method” section, encompassing an introduction to the principles of the diffusion model, deep learning architectures, and data enhancement techniques. The “Experimental studies” section employs two rolling bearing datasets to validate the proposed method. The paper concludes with a summary in the “Conclusion” section.

2. Proposed method

In this section, we describe the basic principles of the diffusion model, introduce the forward process, the reverse process, the posterior distribution of the diffusion model, and how to construct the loss function. In the “Model establishment” chapter, we introduce the neural network structure used in the diffusion model to predict the noise, and the neural network structure used in the bearing data fault diagnosis.

2.1. DDPM

Denoising diffusion probabilistic models akin to other generative models, denoising diffusion probabilistic models facilitate the generation of target data samples by starting with noise sampled from a simple distribution.

The process of Denoising Diffusion Probabilistic Models (DDPMs), which involves iteratively refining a noisy image to generate high-quality samples, is illustrated in Fig. 2. It begins with an image that is progressively corrupted by noise through a series of steps, creating a diffusion process. During training, the model learns to reverse this corruption by denoising the image step-by-step, ultimately reconstructing the original high-quality image from noise. This denoising process is guided by a learned neural network, which helps the model generate accurate samples by gradually improving the image quality with each step.

In diffusion models, probabilistic terms play a central role in describing and controlling the randomness and uncertainty of the data generation process. During the diffusion process, noise is progressively added to the data through a series of probability distributions (usually Gaussian), transforming the original data distribution into one that closely resembles Gaussian noise. In the reverse process, diffusion models use probabilistic terms for denoising, aiming to generate realistic data from pure noise. During training, maximum likelihood estimation is employed to find the probability distribution of the Markov chain in the reverse process, with the goal of minimizing the difference between the generated and real data distributions. Probabilistic terms also influence the model’s parameterization, determining how noise is added and removed at each step. Thus, diffusion models achieve data generation and recovery by controlling these probabilistic terms.

2.1.1. Forward process

In this section, we introduce the forward process of the diffusion model, the forward diffusion stage transforms the clear image into a noisy image by gradually adding Gaussian noise in preparation for the subsequent reverse diffusion stage.

The initial data, X_0 , adheres to distribution $p(X_0)$. The Gaussian noise introduced does not involve trainable parameters; its mean and variance remain constant. The sole determinant of noise intensity is the fixed variance coefficient $\beta_1 \dots \beta_T$.

The entire process is characterized as a fixed Markov chain, where the conditional process is characterized as a fixed Markov chain, where the conditional transition distribution at each step is denoted as $p(X_t|X_{t-1})$, contributing to the overall posterior distribution represented as $p(X_{1:T}|X_0)$.

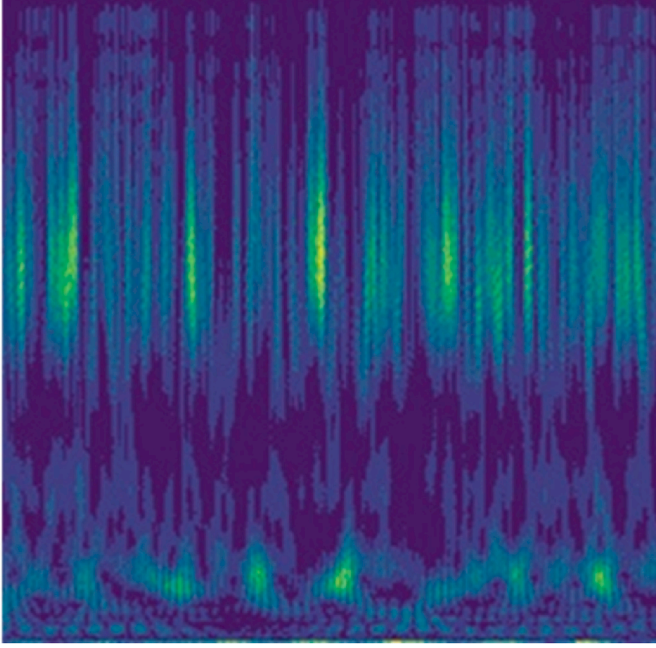


Fig. 3. Vibration data following continuous wavelet transformation.

$$p(X_{1:T}|X_0) = \prod_{t=1}^T p(X_t|X_{t-1}) \quad (1)$$

The transition distribution, $p(X_t|X_0)$, at any given time can be directly derived from the coefficients β_t and the initial state X_0 .

$$p(X_t|X_0) = \mathcal{N}(X_t; \sqrt{\alpha_t}X_0, (1 - \alpha_t)I) \quad (2)$$

where α_t and $\bar{\alpha}_t$ described in equations (3) and (4).

$$\alpha_t = 1 - \beta_t \quad (3)$$

$$\bar{\alpha}_t = \prod_{i=1}^t \alpha_i \quad (4)$$

2.1.2. Reverse process

In this section, we introduce the reverse process of the diffusion model, the reverse diffusion stage trains a noise predictor to predict the noise added to the original image, thereby removing noise from the noisy image and getting a clearer image.

Utilizing neural networks to acquire the transfer distribution.

$$q_\theta(X_{t-1}|X_t) = \mathcal{N}(X_{t-1}; \mu_\theta(X_t, t), \Sigma_\theta(X_t, t)) \quad (5)$$

$$q_\theta(X_{0:T}) = q(X_T) \prod_{t=1}^T q_\theta(X_{t-1}|X_t) \quad (6)$$

The network input, x_t , and time, t , are known. The transfer distribution is characterized by parameters: mean μ_θ and variance Σ_θ . The objective is to solve for θ . The inverse transition distribution, $q_\theta(X_{t-1}|X_t)$, is approximated by the forward posterior distribution, $p(X_{t-1}|X_t, X_0)$.

2.1.3. Objective function

According to Bayes' formula calculated

$$p(X_{t-1}|X_t, X_0) = p(X_t|X_{t-1}, X_0) \frac{p(X_{t-1}, X_0)}{p(X_t, X_0)} \quad (7)$$

$$p(X_{t-1}|X_t, X_0) = \mathcal{N}(X_{t-1}; \tilde{\mu}(x_t, t), \tilde{\beta}_t I) \quad (8)$$

where $\tilde{\beta}_t$ and $\tilde{\mu}(x_t, t)$ described in equations (9) and (10).

$$\tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t \quad (9)$$

$$\tilde{\mu}(x_t, t) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} X_t + \frac{\sqrt{\alpha_{t-1}}\beta_t}{1 - \bar{\alpha}_t} X_0 \quad (10)$$

The diffusion models use maximum likelihood estimation to find the probability distribution of the Markov chain transformation in the process of reverse diffusion, which is the training purpose of the diffusion models. That is, the goal is to maximize the log-likelihood of the model prediction distribution, and to minimize the negative log-likelihood from the point of view of Loss reduction. Its upper bound is determined through variational inference, specifically, the cross-entropy of the joint probability distribution $p(X_{1:T}|X_0)$ for the forward diffusion process and $q_\theta(X_{0:T})$ for the inverse process given X_0 .

Inference-wise, with X_0 given, the KL divergence between the posterior transfer distribution $p(X_{t-1}|X_t, X_0)$ of the forward diffusion process and the inverse transfer distribution $q_\theta(X_{t-1}|X_t)$ is a sequentially iterative and consistently updated Gaussian distribution.

The mean μ and variance Σ of $p(X_{t-1}|X_t, X_0)$ are determined. The variance Σ_θ in the hypothesis $q_\theta(X_{t-1}|X_t)$ is solely linked to the special value of β ; thus, the focus is on learning $\mu_\theta(X_t, t)$.

The loss function simplifies to

$$L_t^{\text{simple}} = E_{t \sim [1, T], X_0, \epsilon_t} \left[\left\| \epsilon_t - \epsilon_\theta \left(\sqrt{\alpha_t} X_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon_t, t \right) \right\|^2 \right] \quad (11)$$

where ϵ_θ represents the predicted value of a neural network with parameters, while ϵ_t denotes the random amount of noise at the current moment.

2.1.4. Model inference

The diffusion model is a process of gradually denoising samples generated by random Gaussian noise to generate new samples. The provided Gaussian noise is the data at the X_t step. Then the noise ϵ_θ predicted by the neural network is gradually denoised, this process will stop at timestep 1. By equation (5) we get $X_{t-1} \sim q_\theta(X_{t-1}|X_t)$, then we can further get the calculation formula from X_t to X_{t-1} , such as equation (12).

$$X_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left[X_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon(X_t, t) \right] + \sigma_t \mathcal{Z} \quad (12)$$

where $\mathcal{Z} \sim \mathcal{N}(0, I)$ is a two-dimensional standard Gaussian distribution. The outlined inference algorithm is depicted in Algorithm 1.

Algorithm1. DDPM algorithm

```

1:  $X_t \sim \mathcal{N}(0, I)$ 
2: for  $t = T, \dots, 1$  do
    $\mathcal{Z} \sim \mathcal{N}(0, I)$  if  $t > 1$ , else  $\mathcal{Z} = 0$ 
    $X_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left[ X_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon(X_t, t) \right] + \sigma_t \mathcal{Z}$ 
3: end for
4: return  $X_0$ 

```

2.2. Model establishment

Initially, our starting point is the assumption that the sensor captures the primary mechanical vibration signal. In the process of data augmentation of a one-dimensional vibration signal using a diffusion model, we find this difficult, so we need to preprocess the one-dimensional vibration signal. Following this, we implement the continuous wavelet transform to convert the acquired vibration signals (shown in Fig. 3). This transformation alters the one-dimensional vibration signals into two-dimensional image signals, preparing them for further processing. Subsequently, the resultant image dataset undergoes partitioning into three distinct segments: the validation dataset, the test

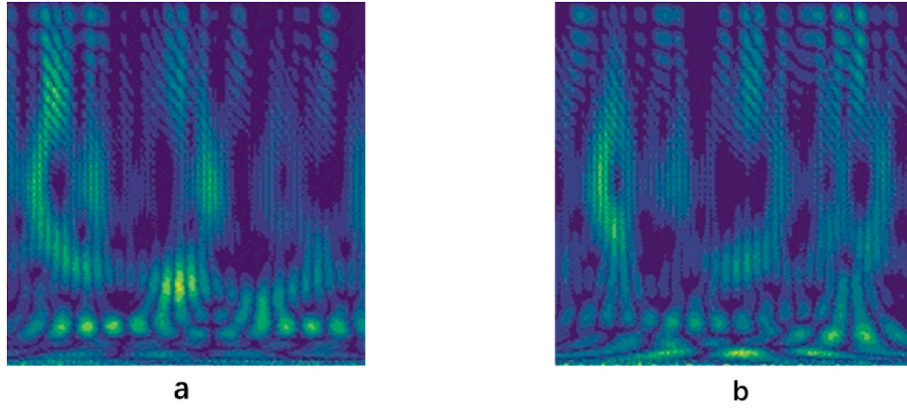


Fig. 4. Figure a represents a two-dimensional image sample from the Paderborn dataset, exhibiting mixed damage of inner and outer circles, while Figure b depicts a DDPM-generated sample sharing the same label.

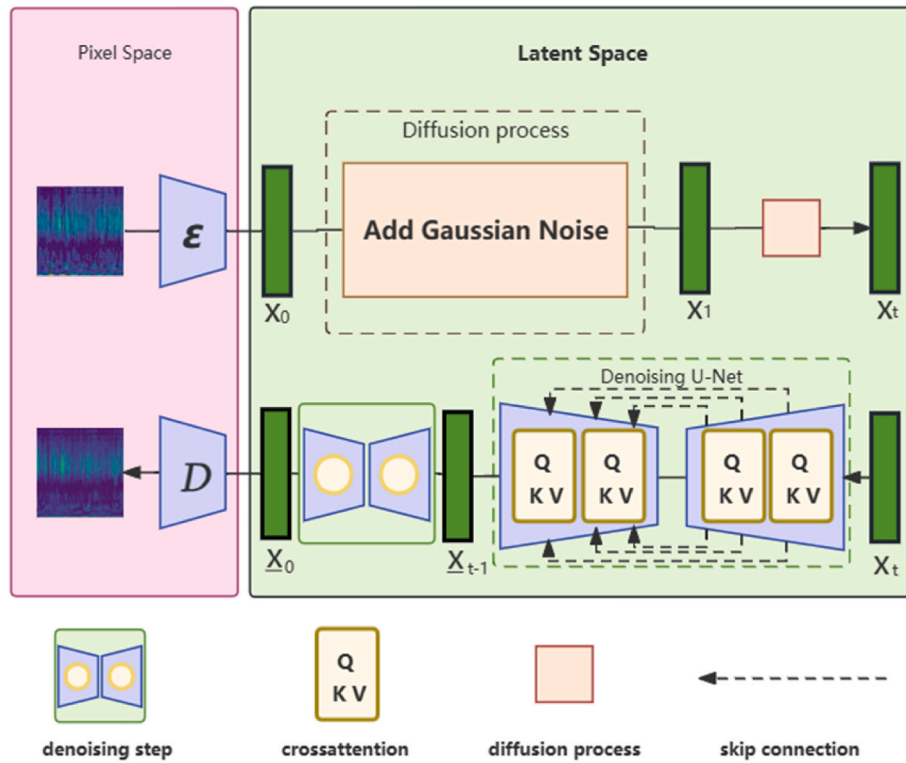


Fig. 5. DDPM structure diagram.

dataset, and the training dataset, serving as input for data enhancement. Subsequent to this partitioning, we proceed to prepare both the training and test datasets for subsequent analysis and modeling.

The data enhancement model utilizes the two-dimensional image derived from the wavelet transformation as its input, whereas the diffusion model employs samples of identical fault types to generate fresh two-dimensional image samples. These novel samples demonstrate similar traits to the initial input, albeit potentially diverging in particulars, as demonstrated in Fig. 4. The potential of employing these freshly generated samples by denoising diffusion probabilistic models (DDPM) for data enhancement will be investigated in forthcoming chapters. Both the improved 2D image samples and the original, unenhanced training samples serve as inputs for the model. The design of the proposed deep learning architecture is subsequently determined, taking into account specific fault diagnosis issues and dataset characteristics.

The diffusion model is employed to generate new training samples,

operating with an epoch of 3000 and a time step of 1000. The U-Net network, well-known for its efficacy in image processing, serves as the denoising model in the DDPM framework. Its primary task involves noise prediction, a critical step laying the groundwork for subsequent processes within the diffusion model. During the encoder operation of the U-Net network, 4–5 pooling layers are employed for continuous resolution reduction through downsampling, acquiring image data across various scales. This progression sees a shift from foundational information like points, lines, and gradients towards more abstract contours and elevations. The network completes feature extraction and integration, enhancing the comprehensiveness of information obtained. Furthermore, U-Net's architecture integrates skip connections, facilitating seamless information flow across layers. These connections enable the network to fuse feature maps from corresponding encoder positions during each level's upsampling. By merging low-level and high-level features, the network preserves high-resolution details from

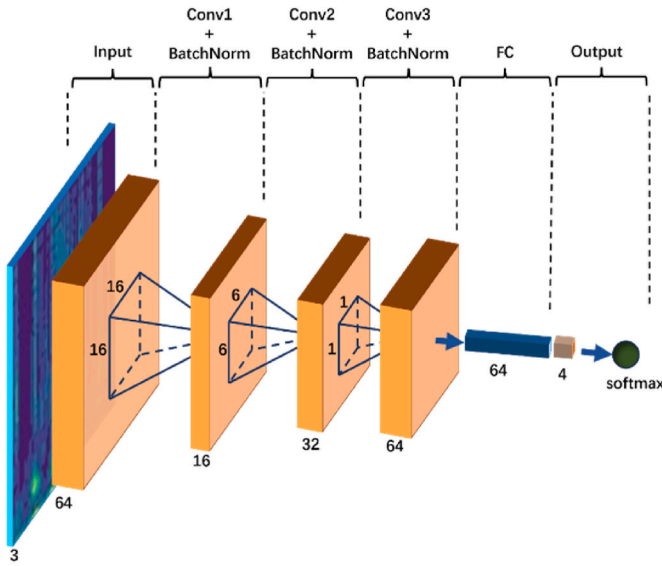


Fig. 6. Convolutional neural network structure diagram.

the latter, thereby enhancing image segmentation accuracy. Cross-layer feature reconstruction augments feature information transmission, leveraging rich details from high-level convolutional layers and maximizing information utilization across the network's layers. A U-Net network without skip connections would have a bottleneck, causing information loss. Typically, the dimension of the middle layer is much smaller than that of the input, contributing to the fuzzy generation seen in Variational auto-encoder (VAE) series. The overall structure of U-Net is straightforward, robust, and efficient, significantly minimizing loss (Liu et al., 2019). The structure of the diffusion model is depicted in Fig. 5.

To initiate the deep learning process, training samples are first transmitted into the proposed neural network. This network consists of multiple layers, each capable of learning the underlying features of the data, which are combined to form a highly abstract representation. At the end of the structure, the learned features are used to classify the fault categories of rolling bearings. The network's weights are updated using backpropagation. The training process is repeated until the network's performance reaches a satisfactory level or the training rounds are completed. After training, test data samples are fed into the trained network, and the accuracy of fault diagnosis is evaluated by calculating the consistency between the predicted outputs and the actual labels. This accuracy rate reflects the network's reliability and effectiveness in identifying faults in practical applications.

The fault classification convolutional neural network, illustrated in Fig. 6, comprises three main sections: the input layer, hidden layers, and output layer. Operating on wavelet images as input, it provides classification and diagnosis results post-training. The hidden layers include 3 convolution layers, 3 pooling layers, and 2 fully connected layers. The convolutional layers consist of 16, 6, and 1 sets of convolutional kernels, respectively, activated by the ReLU function. The final normalized layer connects to a fully connected layer with 64 neurons, further linked to a fully connected layer with either 11 or 4 neurons activated by softmax through a randomly deactivated layer with a parameter of 1. The test model is optimized using the Adam loss optimizer with a learning rate of 0.001. The classification cross-entropy loss function is employed for model loss calculation, with 64 samples inputted for each batch, aligning with test requirements.

3. Experimental studies

In this section, we investigate the impact of three types of data augmentation via an experimental study. Two datasets are employed to

assess the viability of augmentation with the diffusion model on the CWRU dataset, while three enhancement methods were compared on the Paderborn dataset. The network's performance is thoroughly examined, considering various data augmentation techniques, network depths, among other factors. These results are then compared with recent findings on the same dataset, confirming the effectiveness and superiority of the proposed method. For more information on the datasets used, refer to the "Experiments on CWRU Datasets" and the "Experiments on Paderborn Datasets".

All experiments were implemented using the PyTorch platform. Specifically, experiments employing the diffusion model in sample enhancement were carried out on a PC equipped with a GeForce GTX 1080 Ti GPU. The remaining experiments were conducted on a PC with a GeForce GTX 3060 GPU.

3.1. Experiments on CWRU dataset

This section explores data augmentation methods utilizing diffusion models on the CWRU rolling bearing dataset from the Bearing Data Center of Case Western Reserve University.

1. Data introduction

The CWRU (Case Western Reserve University) bearing dataset is a widely used benchmark in fault diagnosis research. It consists of vibration data collected from bearings under various conditions. The data was recorded using accelerometers at different fault sizes and motor loads. The signals are sampled at a high frequency, providing rich time-domain data, which can be transformed into frequency or time-frequency representations for fault detection and classification tasks. The Case Western Reserve University bearing test platform consists of a 2 HP electric motor, a torque sensor/decoder, a power tester, and electronic controllers. The bearing under examination supports the rotating shaft of the motor, with the driving end bearing being SKF6205 and a sampling frequency of 12 KHz and 48 KHz. The fan end bearing is SKF6203, with a sampling frequency of 12 KHz. The motor speed fluctuates between 1730 and 1797 RPM, depending on the applied load (Hendriks et al., 2022).

The vibration signals analyzed in this section are derived from the motor drive end of the test bench, capturing four distinct health states: (1) normal state (H), (2) outer ring failure (OF), (3) inner ring failure (IF), and (4) rolling body failure (BF). All three faults originate from Electrical Discharge Machining (EDM), with fault diameters measuring 7 mils, 14 mils, and 21 mils (1 mil = 0.001 inches). As a result, the dataset encompasses 10 bearing health states under 4 loads, grouping identical health states under different loads into a single class.

2. Experimental settings

In this study, samples from each health state of the driving end bearing, with a sampling frequency of 12 KHz under a single load, are chosen as input for continuous wavelet transform. From all output images, 100 are selected for the training set, while 2500 are chosen for both the validation and test sets, respectively. The two-dimensional convolutional neural network (2D-CNN) is ideal for processing such data due to its ability to effectively capture local spatial features with convolutional filters, detecting patterns like edges and textures. Therefore, a fault classification experiment was conducted using a 2D-CNN, trained with a batch size of 200 for 300 epochs. After training on ten fault conditions, the test dataset achieved an accuracy of 74%.

To assess the viability of the diffusion model for data enhancement, 100 images from the training set are employed as input samples for the diffusion model, generating 500 synthetic vibration images after wavelet transformation. These 500 images, along with the original training set, constitute a new training set, while the validation and test sets remain unaltered. Utilizing a batch size of 200, an epoch number of

Table 1
CWRU dataset generated by DDPM.

	Original data	With 500 generated data
Test accuracy	74.00%	87.60%

300, and employing a two-dimensional convolutional neural network for fault classification, the test accuracy is evaluated.

3. Experimental results

Fault diagnosis on the original data achieved an accuracy of 74%. After data augmentation using the diffusion model, the test accuracy achieved is 87.6%. The test results of the original data and the

augmented data are shown in Table 1, and confusion matrices are shown in Fig. 7.

3.2. Experiments on paderborn dataset

In this section, the data enhancement method based on the diffusion model is investigated using the German Paderborn bearing dataset.

Table 2
Paderborn dataset generated by Gaussian Noise.

	original data	with 40 generated data	with 80 generated data
Data 1	0.798	0.842	0.852
Data 2	0.738	0.803	0.820

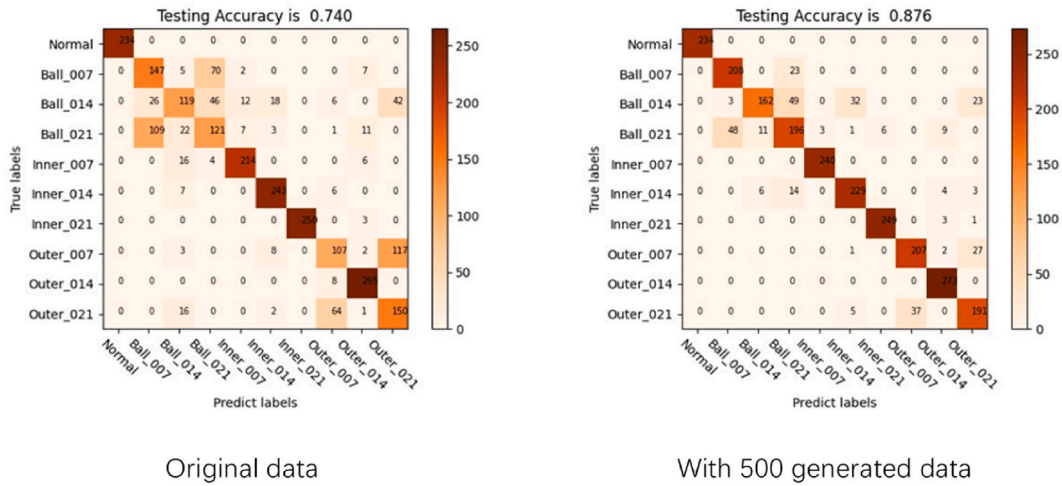


Fig. 7. Confusion matrices of CWRU dataset generated by DDPM.

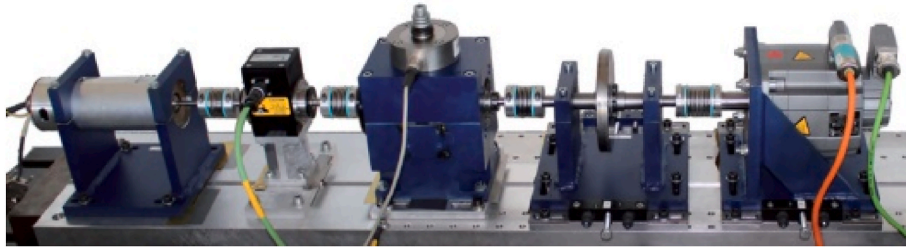


Fig. 8. Paderborn bearing test stand.

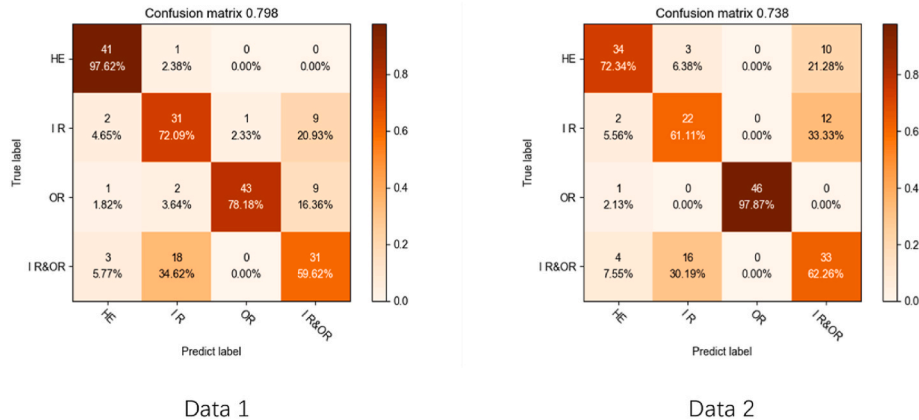


Fig. 9. Confusion matrices of Paderborn dataset.

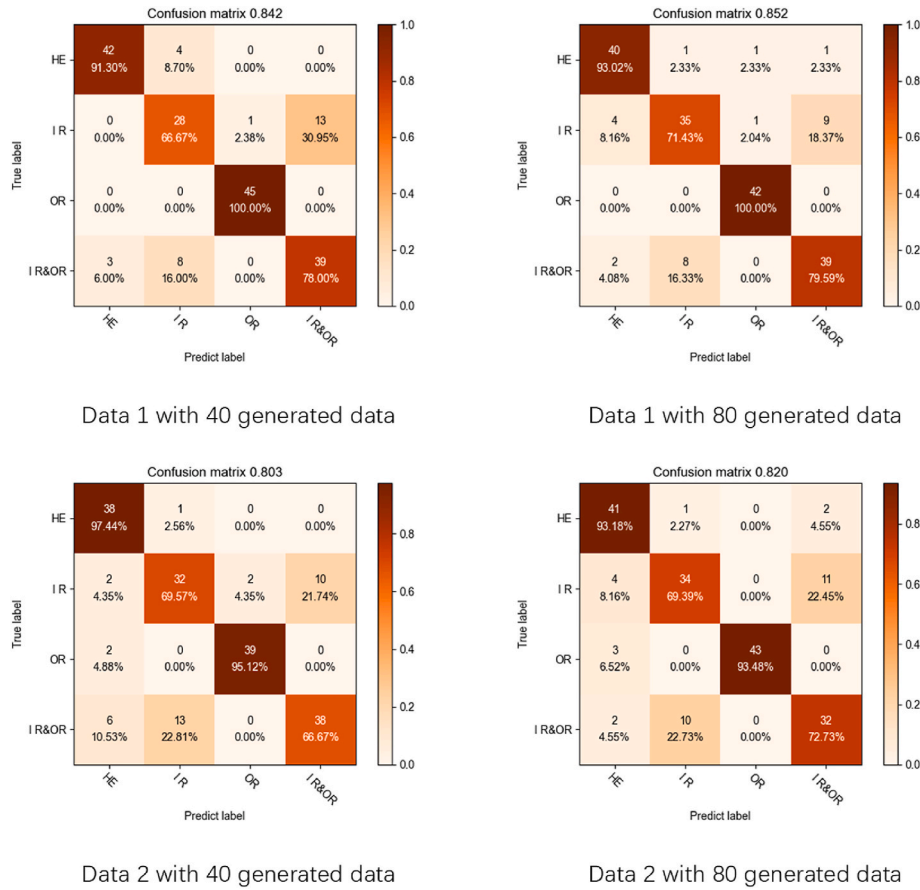


Fig. 10. Confusion matrices of Paderborn dataset generated by Gaussian Noise.

Table 3

The deconvolution structure of the Generator.

No.	Layer Type	Kernel Size/ Stride	Input Size	Output Size	Padding
1	transposed convolution	$4 \times 4/1 \times 1$	100	512	No
2	transposed convolution	$4 \times 4/2 \times 2$	512	256	Yes
3	transposed convolution	$4 \times 4/2 \times 2$	256	128	Yes
4	transposed convolution	$4 \times 4/2 \times 2$	128	64	Yes
5	transposed convolution	$4 \times 4/2 \times 2$	64	100	Yes

Table 4

The deconvolution structure of the Discriminator.

No.	Layer Type	Kernel Size/Stride	Input Size	Output Size	Padding
1	convolution	$4 \times 4/2 \times 2$	100	64	Yes
2	convolution	$4 \times 4/2 \times 2$	64	128	Yes
3	convolution	$4 \times 4/2 \times 2$	128	256	Yes
4	convolution	$4 \times 4/2 \times 2$	256	512	Yes
5	convolution	$4 \times 4/1 \times 1$	512	1	No

Published in 2016 by Paderborn University in Germany, this dataset aims to support the application of data-driven classification methods for monitoring bearing damage in electromechanical drive systems using motor current signals. The dataset comprises numerous vibration signals generated by the bearing test bench, illustrated in Fig. 8.

Table 5

Paderborn dataset generated by DCGAN.

	original data	with 40 generated data	with 80 generated data
Data 1	0.798	0.847	0.825
Data 2	0.738	0.814	0.831

1. Data introduction

The test bench comprises a motor, a torque measuring shaft, a rolling bearing test module, a flywheel, and a load motor. Experimental data are generated by installing ball bearings with various damage types in the bearing test module. The motor current is sampled at a frequency of 64 KHz, the vibration signal at 64 KHz, and mechanical parameters (loading force, loading moment, speed) at 4 KHz. Temperature is sampled at 1Hz. All test bearing models are 6203 rolling bearings. Bearing failures are categorized into artificial and real damage. Artificial damage is primarily induced through electric spark (cracking), drilling (peeling), and electric engraving machine (pitting). Real damage bearings undergo accelerated life testing. Injury types include single point, repetitive, and multiple injuries. Repetitive and multiple injuries are further classified as regular, random, or non-repetitive injuries (Huo et al., 2023).

The dataset comprises 12 artificially damaged bearings (7 with outer ring damage, 5 with inner ring damage), 14 bearings damaged through accelerated life testing (5 with outer ring damage, 6 with inner ring damage, and 2 with compound inner and outer ring damage), and 6 healthy bearings. In addition to vibration signals, the data also includes motor current signals, with the objective of achieving fault diagnosis in rotating machinery through analysis of the motor current signal.

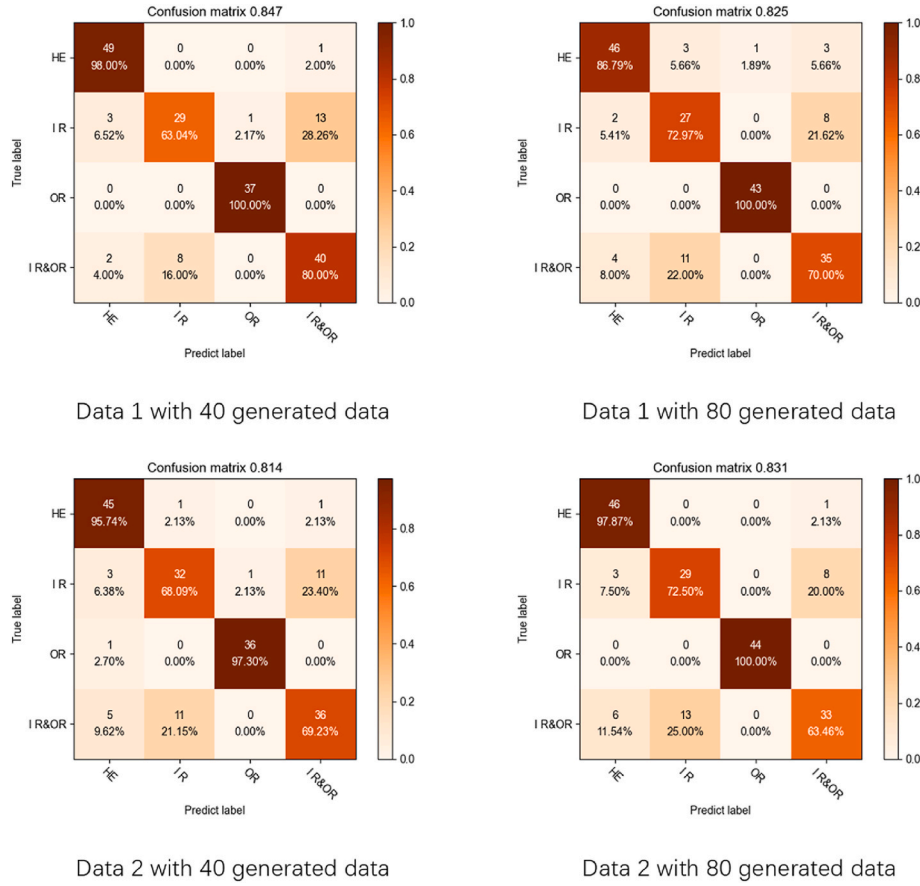


Fig. 11. Confusion matrices of Paderborn dataset generated by DCGAN.

Table 6

Paderborn dataset generated by DDPM.

	original data	with 40 generated data	with 80 generated data
Data 1	0.798	0.896	0.923
Data 2	0.738	0.842	0.863

2. Experimental settings

In this study, samples of 6203 deep groove ball bearings, with a pitch circle diameter of 28.55 mm, were chosen as input for continuous wavelet transform under the conditions of a test speed of 900 revolutions per minute, a loading torque of 0.7N*m, and a radial force of 1000N, considering four health states (inner ring damage, outer ring damage, mixed inner and outer ring damage, and healthy state). From all the resulting images, we will build two training datasets, each containing 40 different wavelet transform images. There are four working conditions: healthy state, inner ring damage, outer ring damage, and mixed inner and outer ring damage, with ten images in each working condition. We named these two different training sets “Data 1” and “Data 2.” Then, two sets of 400 wavelet transform images were selected as the validation and testing sets. The batch size and epoch number for the training and test sets were set to 200 and 300, respectively. Utilizing a two-dimensional convolutional neural network for fault classification.

3. Experimental results

The achieved test accuracy for Data 1 is 79.8%, and for Data 2, it is 73.8%. The accuracy confusion matrices of Data 1 and Data 2 are shown in Fig. 9.

3.2.1. Data augmentation based on Gaussian noise

This section explores the impact of data augmentation through the addition of Gaussian noise. By adding a Gaussian noise to the sample, we obtain the augmented signal sample (Li et al., 2020). Gaussian noise is a popular method for data augmentation in fault diagnosis, offering both benefits and drawbacks. Its advantages include ease of generation and application, enhanced model robustness by reducing sensitivity to minor data variations, and prevention of overfitting by encouraging the learning of generalized features. However, Gaussian noise also has limitations: it introduces random variations that may not accurately represent diverse or realistic fault conditions, potentially obscuring subtle fault features. Additionally, excessive noise can lead to overly generalized models that miss critical fault details. The target for adding Gaussian noise is 40 vibrational images from the original training set. Noise, generated from a normal distribution, is introduced. The mean of the distribution is 0 and the variance of the distribution is 0.01.

An additional 40 and 80 noisy images are generated and incorporated into the original 40 vibrating images, forming two new training sets, while the validation set and test set remain unaltered. The training and test sets have a batch size of 200 and an epoch number of 300. Utilizing a two-dimensional convolutional neural network for fault classification, test accuracies of 84.2% and 85.2% are achieved for Data 1, respectively. Test accuracies of 80.3% and 82% are achieved for Data 2, respectively. The test results of the original data and the augmented data are shown in Table 2, and confusion matrices are shown in Fig. 10.

3.2.2. Data augmentation based on DCGAN

This section explores the impact of data augmentation through the addition of DCGAN-generated images. Deep Convolutional Generative Adversarial Network (DCGAN), based on traditional Generative Adversarial Networks (GAN), incorporates convolutional neural networks into

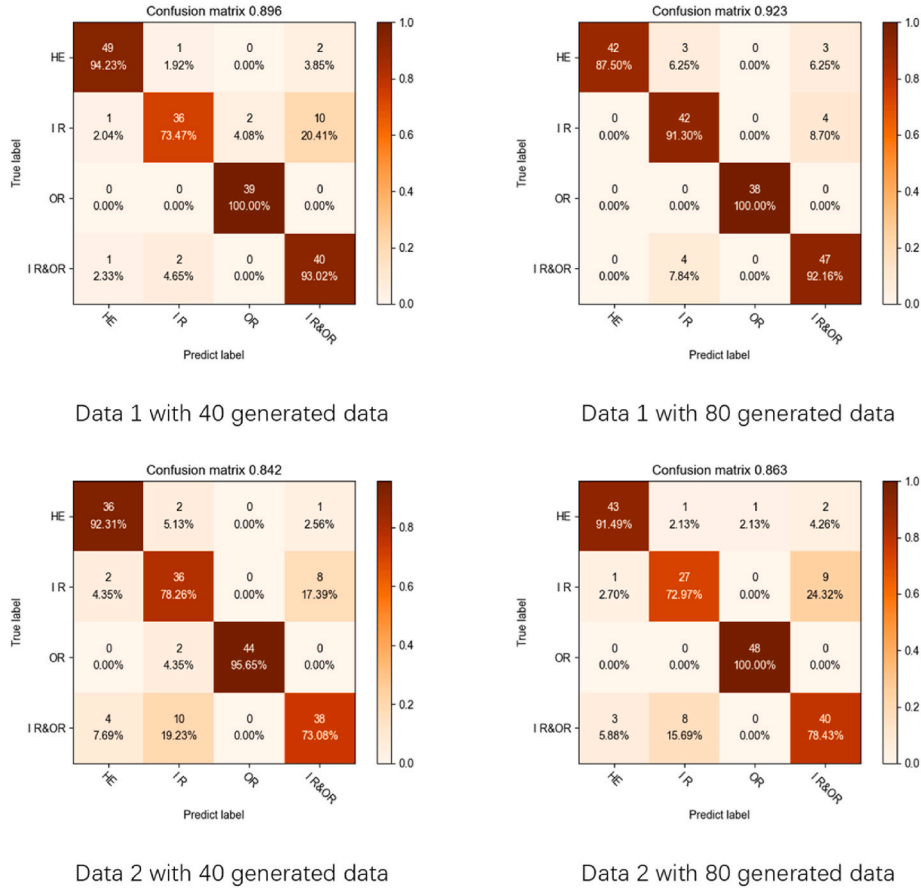


Fig. 12. Confusion matrices of Paderborn dataset generated by DDPM.

Table 7
Comparison of 3 methods on Data 1.

Data 1	original data	with 40 generated data	with 80 generated data
Gaussian Noise	0.798	0.842	0.852
DCGAN	0.798	0.847	0.825
DDPM	0.798	0.896	0.923

Table 8
Comparison of 3 methods on Data 2.

Data 2	original data	with 40 generated data	with 80 generated data
Gaussian Noise	0.738	0.803	0.820
DCGAN	0.738	0.814	0.831
DDPM	0.738	0.842	0.863

the training to enhance the effect of generative networks (Luo et al., 2021). DCGAN takes 40 vibration images from the original training set as input. The feature map size in both the Generator and Discriminator is set to 64, with training epochs totaling 10,000. The learning rate for optimizers is set at 0.0005, and the hyperparameters for Adam optimizers is 0.5. The deconvolutional structure of the Generator is outlined in Table 3, while the deconvolutional structure of the Discriminator is presented in Table 4.

An additional 40 and 80 new images are generated and incorporated into the original 40 motion images, forming 2 new training sets, while the validation set and test set remain unaltered. The training set and test

set have a batch size of 200 and an epoch number of 300. Utilizing a two-dimensional convolutional neural network for fault classification, Data 1 achieves test accuracies of 84.7% and 82.5%, respectively. Data 2 achieves test accuracies of 81.4% and 83.1%, respectively. The test results of the original data and the augmented data are shown in Table 5, and confusion matrices are shown in Fig. 11.

3.2.3. Data augmentation based on DDPM

This section explores the impact of data augmentation through the addition of diffusion model-generated images. The input samples for the diffusion model consist of 40 vibration images from the original training set. The diffusion model employed aligns with the structural setup outlined in Section 3.3.1.

An additional 40 and 80 new images are generated and incorporated into the original 40 motion images, forming 2 new training sets, while the validation set and test set remain unaltered. The training set and test set have a batch size of 200 and an epoch number of 300. Utilizing a two-dimensional convolutional neural network for fault classification, Data 1 achieves test accuracies of 89.6% and 92.3%, respectively. For Data 2, the test accuracies were 84.2% and 86.3%, respectively. The test results of the original data and the augmented data are shown in Table 6, and confusion matrices are shown in Fig. 12.

4. Conclusions

In this paper, we present a method for enhancing data utilizing a diffusion model to aid in the diagnosis of bearing faults in scenarios with limited sample availability. We begin by elucidating the foundational principles of denoising diffusion probabilistic models. Following this, we detail the process of the proposed methodology, where vibration signals

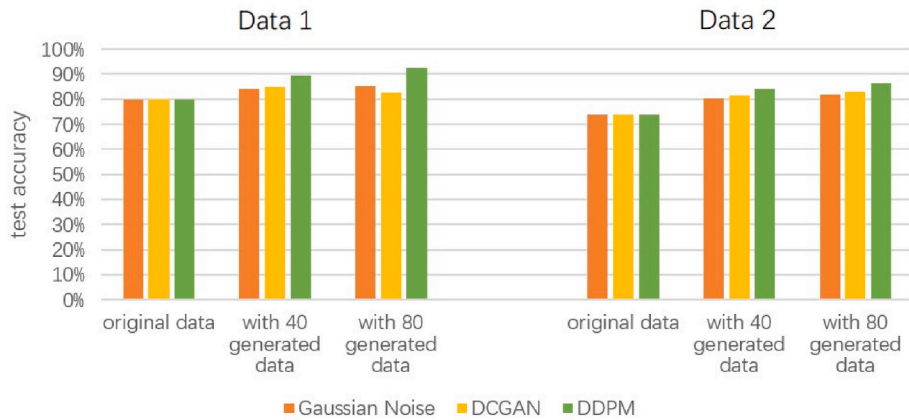


Fig. 13. Comparison of 3 methods on Paderborn dataset.

undergo conversion into image signals via continuous wavelet transform. This conversion facilitates the creation of novel samples for data augmentation utilizing the diffusion model. Experiments conducted on the CWRU and Paderborn datasets led to two conclusions.

1. The diffusion model demonstrates the capability to generate high-quality images with features akin to the original samples. It accurately produces fault data wavelet image samples, especially beneficial under conditions of a small sample database, effectively expanding and enhancing the dataset. In the Case Western Reserve University bearing dataset, data augmentation using the diffusion model achieved an accuracy of 87.6%, notably surpassing the 74% accuracy of the original samples.
2. Upon comparing and validating the Paderborn University bearing dataset, it becomes evident that the wavelet image samples generated through the diffusion model exhibit high authenticity. The accuracy rates of Data 1 reached 89.6% and 92.3% respectively for the addition of 40 and 80 samples, while Data 2 achieved accuracy rates of 84.2% and 86.3% respectively for the same sample additions, showcasing superior performance compared to data enhancement methods such as adding Gaussian noise and DCGAN. The test results of Data 1 and Data 2 are shown in Tables 7 and 8, and the bar chart is depicted in Fig. 13.

Diffusion models generate data by progressively adding and removing noise in a controlled manner, enabling them to learn complex data distributions in detail. This results in high-quality, realistic samples that more accurately reflect the variability of real-world data compared to conventional augmentation techniques like flipping, rotation, or adding simple noise. They can produce a wide range of data variations, including subtle changes difficult to achieve with traditional methods. This diversity enhances the robustness and generalization of machine learning models trained on augmented data. Additionally, diffusion models excel at denoising, making them particularly effective in scenarios with noisy or incomplete data. This capability not only produces clean and accurate samples but also improves model performance in noisy environments, where traditional augmentation methods often fall short.

Overall, using Denoising Diffusion Probabilistic Models (DDPM) for data augmentation has both advantages and disadvantages. DDPM excels in training stability, sample quality, mode coverage, and robustness to noisy data compared to DCGAN and adding Gaussian noise methods, resulting in superior data augmentation outcomes. However, the original diffusion model faces challenges in generating high-quality one-dimensional time-series data, leading us to use two-dimensional images as model inputs, which adds complexity to the process. Additionally, DDPM requires substantial computational resources and has slower

sample generation speeds, although these issues may be addressed with future advancements.

Future research on data augmentation in fault diagnosis will likely focus on generating more realistic synthetic fault data, improving generalization across different machines. This includes integrating physics-based models with machine learning, enhancing transfer learning for cross-domain applications, and using reinforcement learning to create dynamic augmentation strategies. Additionally, explainable AI could boost the reliability of augmented data. Other solutions to the scarcity of faulty data include Few-Shot Learning, Semi-Supervised Learning, Anomaly Detection, and Transfer Learning (Yang et al., 2024c), which can reduce the need for large labeled datasets.

CRedit authorship contribution statement

Pengcheng Zhao: Formal analysis, Data curation. **Wei Zhang:** Methodology. **Xiaoshan Cao:** Investigation. **Xiang Li:** Resources, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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